

05300
TERMS/LANDING MINIMA DATA
IFR LANDING MINIMA

The United States Standard for Terminal Instrument Procedures (TERPS) is the approved criteria for formulating instrument approach procedures. Landing minima are established for six aircraft approach categories (ABCE and COPTER). In the absence of COPTER MINIMA, helicopters may use the CAT A minimums of other procedures. The standard format for RNAV minima and landing minima portrayal follows:

RNAV (GPS) MINIMA

CATEGORY	A	B	C	D
LPV DA		1540/24	258 (300-1/2)	
INNAV/NAV		1600/24	318 (400-1/2)	1600/40 318 (400-3/4)
DA				
INNAV MDA	1840/24	558 (600-1/2)	1840/50 558 (600-1)	1840/60 558 (600-1 1/4)
CIRCLING	1840-1	545 (600-1)	1840-1 1/2 545 (600-1 1/2)	1860-2 565 (600-2)

NOTE: The **W** symbol indicates outages of the WAAS vertical guidance may occur **daily** at this location due to initial system limitations. WAAS NOTAMS for vertical outages are not provided for this approach. Use INAV minima for flight planning at these locations, whether as a destination or alternate. For flight operations at these locations, when the WAAS outages indicate that LNAV/VNAV or LPV service is available, then vertical guidance may be used to complete the approach using the displayed level of service. Should an outage occur during the procedure, reversion to LNAV minima may be required. As the WAAS coverage is expanded, the **W** will be removed.

RNAV minimums are dependent on navigation equipment capability, as stated in the applicable AFM, AFMS, or other FAA approved document and as outlined below

GLS (Global Navigation Satellite System (GNSS) Landing System)

The GLS (NA) minima line will be removed from existing RNAV (GPS) approach charts when LPV minima is published.

LPV (An Approach Procedure with Vertical Guidance (APV) based on WAAS lateral and vertical guidance)

Must have WAAS avionics approved for LPV approach.

LNNAV/VNAV (Lateral navigation/Vertical navigation)

Must have either:

g) WAS avionic

- WAAS avionics approved for LNAV/VNAV approach, or
- A certified Baro-VNAV system with an IFR approach approved GPS, or
- A certified Baro-VNAV system with an IFR approach approved WAAS, or
- An approach certified RNP-0.3 system with barometric vertical guidance (Baro-VNAV). Other RNAV systems require special approval.

Other NNAV systems require spectral approval.

NOTES:

1. LNAV-VNAV minima not applicable for Baro-VNAV equipment if chart is annotated "Baro-VNAV NA" or when below the minimum published temperature, e.g., Baro-VNAV NA below -7°C (20°F).

- "DME/DME based RNP-0.3 systems may be used only when a chart note indicates DME/DME availability; e.g., the minimum published temperature, e.g., baro-VNAV (NA below -1° C (-1° F)).

LNAV (Lateral navigation)

NOTE: DME/DME based RNP-0.3 systems may be used only when a chart note indicates DME/DME availability; e.g., "DME/DME RNP-0.3 Authorized." Specific DME facilities may be required; e.g., "DME/DME RNP-0.3 Authorized. ABC XY7 required."

LANDING MINIMA FORMAT

In this example airport elevation is 1179, and runway touchdown zone elevation is 1152.

CATEGORY	Aircraft Approach Category			
	A	B	C	D
S-ILS 27	1352/24		200	
S-LOC 27	1440/24	288	(300-1/2)	1440/50 288 (300-1)
CIRCLING	1540-1 361 (1400-1)	1640-1 461 (500-1)	1640-1 1/2 461 (500-1 1/2)	1740-2 561 (600-2)

DA: Visibility (RVR 100's of feet)

HAT

All minima in parentheses not applicable to Civil Pilots.

Military Pilots refer to appropriate regulations.

See Table in Supplement

Strat-high in ILS to Runway 27

Strat-high with Glide Slope Inoperative or not used to Runway 27

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Figure 1-20. *Terms/landing minima data.*

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COPTER MINIMA ONLY

CATEGORY	COPTER
H-176°	680-½ 363 (400-½)

Copter Approach Direction

Height of MDA/DA Above Landing Area (HALL)

No circling minimums are provided

RADAR MINIMA

	PAR (c)	10	2.5°/42/1000	ABCDE	195/16	100	(100-¼)	Velocity (RVR 100's of feet)
	(d)	28	2.5°/48/1068	ABCDE	187/16	100	(100-¾)	
ASR		10		ABC	580/40	463	(500-¾)	D
				E	580/60	463	(500-1¼)	
		28		AB	600/50	513	(600-1)	C
				DE	600-½	513	(600-1½)	
CLR (b)		10		AB	580-¼	463	(500-1¼)	C
		28		AB	600-¼	503	(600-1¼)	C
		10, 28		DE	660-2	563	(600-2)	
								560/50 463 [500-1]
								600/60 513 [600-1¼]
								580-½ 463 [500-1½]
								600-½ 503 [600-1½]

All minima step in parentheses not applicable to Civil Pilots. Military Pilots refer to appropriate regulations.

Visibility in Statute Miles

Radar Minima:

1. Minima shown are the lowest permitted by established criteria. Pilots should consult applicable directives for their category of aircraft.
2. The circling MDA and weather minima to be used are those for the runway to which the final approach is flown - not the landing runway. In the above RADAR MINIMA example, a category C aircraft flying a radar approach to runway 10, circling to land on runway 28, must use an MDA of 560 feet with weather minima of 500-1½.

- A** Alternate Minimums not standard. Civil users refer to tabulation. USA/USN/USAF pilots refer to appropriate regulations.
- NA** Alternate minimums are Not Authorized due to unmonitored facility or absence of weather reporting service.
- T** Take-off Minimums not standard and/or Departure Procedures are published. Refer to tabulation.

AIRCRAFT APPROACH CATEGORIES

Aircraft approach category indicates a grouping of aircraft based on a speed of VREF, if specified, or, if VREF not specified, 1.3 VSO at the maximum certificated landing weight. VREF, VSO, and the maximum certificated landing weight are those values as established for the aircraft by the certification authority of the country of registry. Helicopters are Category A aircraft. An aircraft shall fit in only one category. However, if it is necessary to operate at a speed in excess of the upper limit of the speed range for an aircraft's category, the minimums for that category for that speed shall be used. For example, on an approach where this into Category B, but is circling to land at a speed of 145 knots, shall use the approach Category D minimums. As an additional example, a Category A airplane (for helicopter) which is operating at 130 knots on a straight-in approach shall use the approach Category C minimums. See [Category limits](#).

MANEUVERING TABLE

Approach Category	A	B	C	D	E
Speed (Knots)	0-90	91-120	121-140	141-165	Abv 165

Comparable Values of RVR and Visibility

The following table shall be used for converting RVR to ground or flight visibility. For converting RVR values that fall between listed values, use the next higher RVR value; do not interpolate. For example, when converting 1800 RVR, use 2400 RVR with the resultant visibility of 1/2 mile.

RVR (feet)	Visibility (statute miles)	RVR (feet)	Visibility (statute miles)
1600	$\frac{1}{4}$	4500	$\frac{3}{8}$
2400	$\frac{1}{2}$	5000	1
3200	$\frac{5}{8}$	6000	$1\frac{1}{4}$
4000	$\frac{3}{4}$		

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